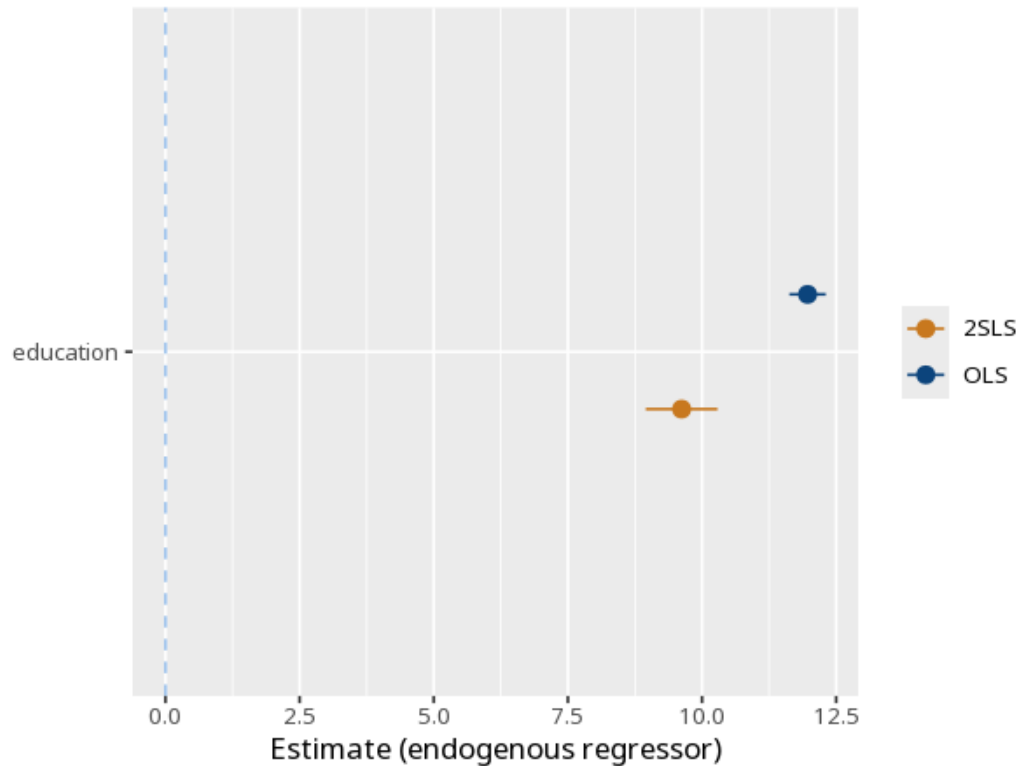


1 Instrumental variables (IV / 2SLS)

Endogenous	Instrument	Coef.	SE	Partial F	p
education	mother_education	0.54	0.02	507.83	<.001
	OLS	2SLS			
(Intercept)	207.67 (2.49) [202.77, 212.57]	240.20 (4.71) [230.92, 249.49]			
education	11.97 (0.17) [11.63, 12.31]	9.62 (0.34) [8.95, 10.29]			
experience	0.62 (0.06) [0.51, 0.72]	0.63 (0.07) [0.50, 0.76]			
Num.Obs.		250			
RMSE		7.52			
F		447.72			
Weak-instrument F = 507.83, p < .001 (rule of thumb: F > 10)					
Wu-Hausman = 1029.75, p < .001					
Sargan: — (just-identified)					



Check the first-stage F first (rule of thumb > 10 = strong instruments). Then read the 2SLS B for the endogenous predictor as the causal effect, with p/CI. This causal reading rests on the instrument being valid - relevant and affecting the outcome only through the endogenous predictor (the exclusion restriction). That is an assumption you must justify on theory; the Sargan test only checks over-identification (extra instruments), not validity, and is unavailable for just-identified models. Method: R ivreg / AER package; weak-instrument F (Stock & Yogo, 2005). Your significance threshold (α) is 0.05.

The 2SLS estimate for education was $B=9.62$, 95% CI [8.95, 10.29], $p < .001$ (first-stage $F=507.83$).

Statistical basis: Two-stage least squares (2SLS) estimation (Fox J, Kleiber C, Zeileis A (2024). ivreg: Instrumental-Variables Regression by 2SLS, 2SM, or 2SMM, with Diagnostics. R package.); Weak-instrument first-stage F rule of thumb (Stock, J. H., Yogo, M. (2005). "Testing for Weak Instruments in Linear IV Regression." In Identification and Inference for Econometric Models (Andrews & Stock, eds.), Cambridge University Press, 80-108.); Heteroscedasticity-robust standard errors (Zeileis A, Köll S, Graham N (2020). "Various Versatile Variances: An Object-Oriented Implementation of Clustered Covariances in R." Journal of Statistical Software, 95(1), 1-36.); Wu-Hausman endogeneity test (Wu, D. M. (1973). "Alternative tests of independence between stochastic regressors and disturbances." Econometrica, 41(4), 733-750.); Sargan over-

identification test (Sargan, J. D. (1958). "The estimation of economic relationships using instrumental variables." *Econometrica*, 26(3), 393-415.).